

Finding Outliers in Subspaces

- ❑ Search for outliers in various subspaces
- ❑ If an object is found to be an outlier in a subspace of much lower dimensionality, the subspace provides critical information for interpreting why and to what extent the object is an outlier
- ❑ Example: Alice, is an outlier in a lower-dimensional subspace that contains the dimensions average transaction amount and purchase frequency, such that her average transaction amount is substantially larger than the majority of the customers, and her purchase frequency is dramatically lower

A Grid-Based Subspace Outlier Detection Method

- ❑ Consider projections of the data onto various subspaces
- ❑ If an area in a subspace has a density that is much lower than average, then the area may contain outliers
- ❑ Discretize the data into a grid in an equal-depth way – each dimension is partitioned into ϕ equal-depth ranges, where each range contains a fraction, $f = \frac{1}{\phi}$, of the objects
 - ❑ Data along different dimensions may have different localities
 - ❑ An equal-width partitioning of the space may not be able to reflect such differences in locality

Measuring Sparsity of Regions in Subspaces

- ❑ Consider a k -dimensional cube formed by k ranges on k dimensions
- ❑ Suppose the data set contains n objects
- ❑ If the objects are independently distributed, the expected number of objects falling into a k -dimensional region is $\left(\frac{1}{\phi}\right)^k n = f^k n$, and the standard deviation is $\sqrt{f^k(1 - f^k)n}$
- ❑ Suppose a specific k -dimensional cube C has $n(C)$ objects, define the sparsity coefficient of C as
$$S(C) = \frac{n(C) - f^k n}{\sqrt{f^k(1 - f^k)n}}$$

Detecting Outliers

- ❑ If $S(C) < 0$, then C contains fewer objects than expected
- ❑ The smaller the value of $S(C)$, the sparser C is and the more likely the objects in C are outliers in the subspace
- ❑ By assuming $S(C)$ follows a normal distribution, we can determine the significance level for an object being an outlier